

Reasoning with Uncertainty

The Doorbell Problem

- The doorbell rang at 12'0 clock in the midnight
 - Was someone there at the door?
 - Did Karim wake up?
- Prop1: AtDoor (x) $\longrightarrow$ Doorbell
- Prop2: Doorbell $\longrightarrow$ wake (Karim)

Reasoning about Doorbell

- Given Doorbell, can we say
- $\text{AtDoor}(x)$, because $\text{AtDoor}(x) \longrightarrow \text{Doorbell}$?
- Abductive reasoning
- But no, the doorbell might start ringing due to some other reason e.g.

Reasoning about Doorbell -II

- Given Doorbell, can we say wake(Karim), because Doorbell $\longrightarrow$ Wake (Karim)
- Deductive reasoning
- Yes, only IF Proposition 2 is always true
- However, in general Karim may not always wakeup, even if the bell rings

Reasoning about Doorbell-III

- Therefore we can't answer either of the questions with certainty
- Proposition 1 is incomplete. Modifying it as
- Does not help because the list of possible causes on the left is huge (infinite???)
- Proposition 2 is often true, but not a tautology

Any way out?

- However problems like that of the doorbell are very common in real life
- In AI, we often need to reason under such circumstances
- We solve it by proper modeling of uncertainty and impreciseness, and developing appropriate reasoning techniques

Sources of uncertainty - I

- Implications may be weak
- Imprecise language like often, rarely, sometimes
- Need to qualify those in terms of frequencies
- Need to design rules for reasoning these frequencies

Sources of uncertainty -II

- Precise information may be too complex
- Too many antecedents or consequents
- Incomplete knowledge
- We may not know or guess all the possible antecedents or consequents
- The bell ring due to some other reasons

Sources of uncertainty -III

- Conflicting information
 - Experts often provide conflicting information
- Propagation of uncertainties
 - In absence of interdependencies propagation of uncertain knowledge increase the uncertainty of the conclusion

Uncertainty -II

Dealing with Uncertainty

- Expressiveness
 - Can concepts used by humans be represented adequately
 - Can the confidence of experts in their decisions be expressed
- Comprehensibility
 - Representation of uncertainty
 - Utilisation in reasoning method
- Correctness
 - Probabilities, relevance ranking, long inference chain
- Computational complexity

Sources of uncertainty

- Data
 - Missing data, unreliable, ambiguous, imprecise representation, inconsistent, subjective, derived from defaults
- Expert knowledge
 - Inconsistency between different experts
 - Plausibility
 - Best guess of expert
 - Quality
 - Causal knowledge- deep understanding
 - Statistical association - observations

- Knowledge representation
 - Restricted model of the real system
 - Limited expressiveness of the representation mechanism
- Inference process
 - Deductive
 - The derived result is formally correct but wrong in the real system
 - Inductive
 - New conclusions are not well-founded
 - Unsound reasoning methods

Uncertainty in individual rules

- Individual rules
 - Errors
 - Domain errors
 - Representation errors
 - Inappropriate application of rules
 - Likelihood of evidence
 - For each premise
 - For the conclusion
 - Combination of evidence from multiple premises

Uncertainty and multiple rules

- Conflict resolution
 - If multiple rules are applicable , which is selected
- Compatibility
 - Contradictions between rules
 - Subsumption
 - One rule is more general version of another one
 - Missing rules
 - Data fusion
 - Integration of data from multiple sources

Basics of probability theory

- Mathematical approach for processing uncertain information
 - Sample space set
 - Collection of all possible events
 - Can be discrete or continuous
 - Probability number
 - Likelihood of an event to occur
 - Non-negative value in $[0,1]$
 - Total probability of the sample space

- For mutually exclusive events, the probability for at least one of them is the sum of their individual probabilities
- Experimental probability
 - Based on the frequency of events
- Subjective probability
 - Based on expert assessment

Compound probabilities

- Describes independent events
 - Do not affect each other in any way
- Joint probability of two independent events

Union probability of two
independent events

Basic axioms of probability

Conditional probabilities

- Describes dependent events
 - Affect each other in some way
- Conditional probability of event

Probability prior vs conditional

- Prior probabilities of events A and B is $P(A)$ and $P(B)$
- But, if we know A for sure, then probability of B might change. This is represented as the conditional probability $P(B|A)$

Probability of Baye's Rule

Uncertainty -III

Probability – independent & joint

- Events A and B are independent if
 - $P(A|B) = P(A)$, $P(B|A) = P(B)$
 - Therefore, $P(A \wedge B) = P(A) * P(B)$
- Joint probability
- example

Advantages and problems of probabilities

- Advantages
 - Formal foundation
 - Reflection of reality (a posteriori)
- Problems
 - May be inappropriate
 - The future is not always similar to the past
 - Inexact or incorrect
 - Especially for subjective probability
 - Knowledge may be represented implicitly

Bayesian Approach

- Derive the probability of a cause given a symptom
- Has gained importance recently due to advances in efficiency
 - More computational power available
 - Better method
- Especially useful in diagnostic system
 - medicine
- Inverse or a posteriori probability
 - Inverse to conditional probability of an earlier event given that a later one occurred

Extended Baye's Theorem

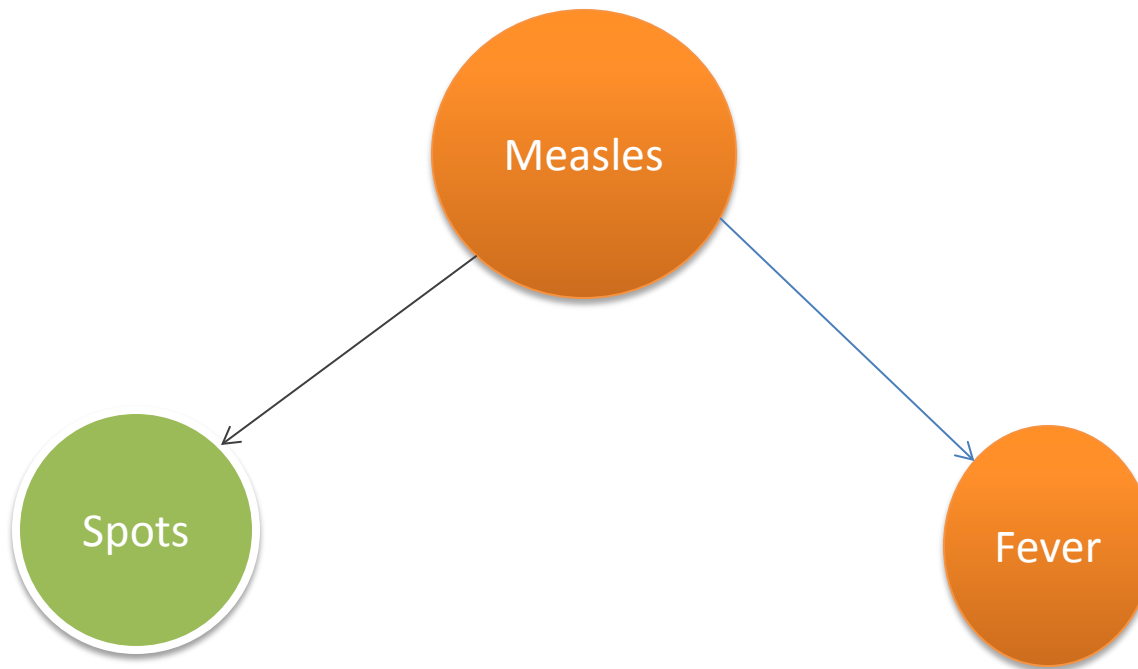
- $P(H_i|E)$ = probability that hypothesis H_i is true given evidence E
- $P(E|H_i)$ = probability that we will always observe evidence E given that hypothesis H_i is true
- $P(H_i)$ = the a priori probability that hypothesis H_i is true in the absence of any specific evidence
- K is the number of possible hypothesis
- Write the equation

Measles is an infectious viral disease causing fever and a red rash on the skin

S: patient has spots

M: patient has measles

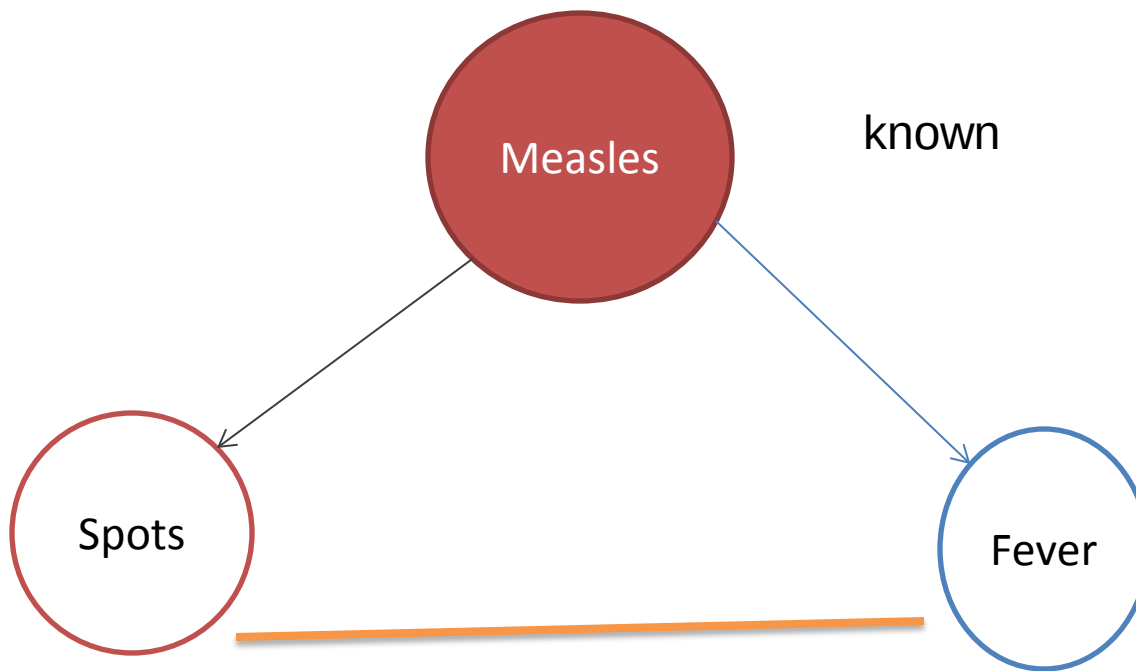
F: patient has high fever



Spots: evidence for measles and also for fever

Measles is an infectious viral disease causing fever and a red rash on the skin

S: patient has spots
M: patient has measles
F: patient has high fever



Additional evidence

Increase belief in favor of fever

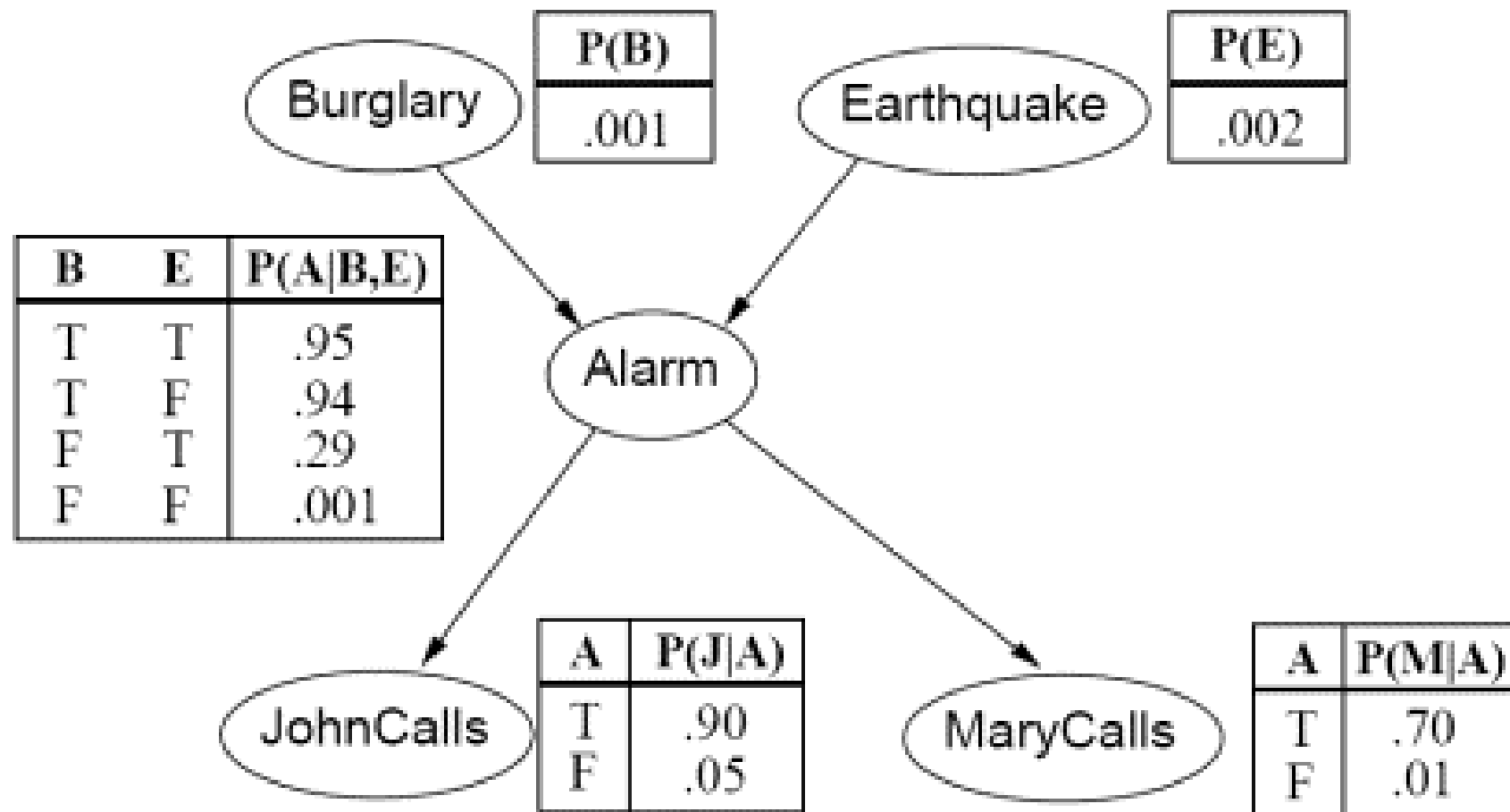
Spots: evidence for measles and also for fever

- If both spots and fever are present they provide evidence in favor of measles
- But they are not independent!!!
- Hence, we can't simply sum their effects but take conditional probabilities of their conjunctions into account

- In general, given a prior body of evidence e and some new observation E , compute
- $P(H|E,e) = P(H|E).P(e|E,H)/P(e/E)$
- In a real world the size of joint probabilities that we need to compute grows as 2^n if there are n propositions to be considered

Belief Network

- Belief network is a representation of causal connection between different events with associated conditional probabilities
- Nodes represent events
- Directed edges from node A to node B represents the fact that event B may be an effect of event A
- Conditional probabilities: for a node is provided for each possible value of its parents



An example (BN)

- Burglar alarm at home
 - Fairly reliable at detecting burglary
 - Responds at times to minor earthquake
- Two neighbors, on hearing alarm, call police
 - Jhon always calls when the alarm rings, but sometimes he confuses thye telephone ring with the alarm and calls then too
 - Mary listens to loud music and sometimes misses the alarm altogether

Some properties of BN

- A BN is a directed acyclic graph (DAG)
- Nodes which have no parents have prior probabilities
- All other nodes have conditional probabilities
- In general there can be multiple directed paths between a given pair of nodes
- A BN in which there is only one directed path between any pair of nodes is called a polytree

Poly-trees

- A polytree is a graph where there is at most one undirected path between any two pair of nodes
- We consider inferencing only for BNs which are poly-trees

Example contd

- What is the probability that the alarm has sounded but neither a burglary nor earthquake has occurred and both Mary and Jhon call
- $P(\neg E \wedge \neg B \wedge A \wedge J \wedge M)$
- $P(J|A)P(M|A)P(A|\neg E \wedge \neg B)P(\neg E)P(\neg B)$
- 0.9X.....

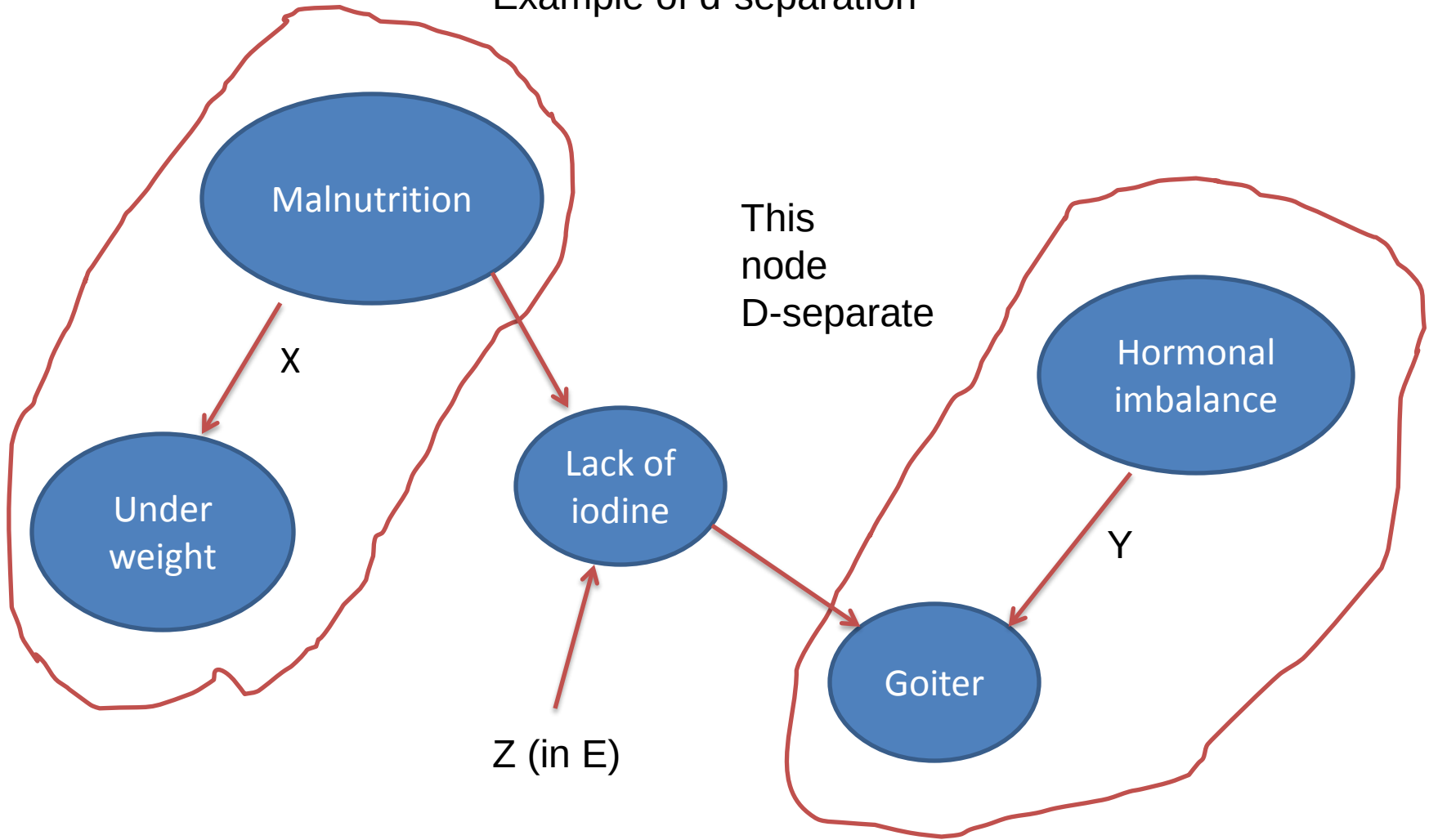
More on conditional independence

- However, if information about the parent nodes is not known, then a node may not be conditionally independent from its ancestors, children or siblings
- For example, if it is not known whether the alarm sounded or not, the event J (Jhon Call) is not independent of the events B (Burglary) or M (mary Calls)
- If we know that B or M true then probability of J increases

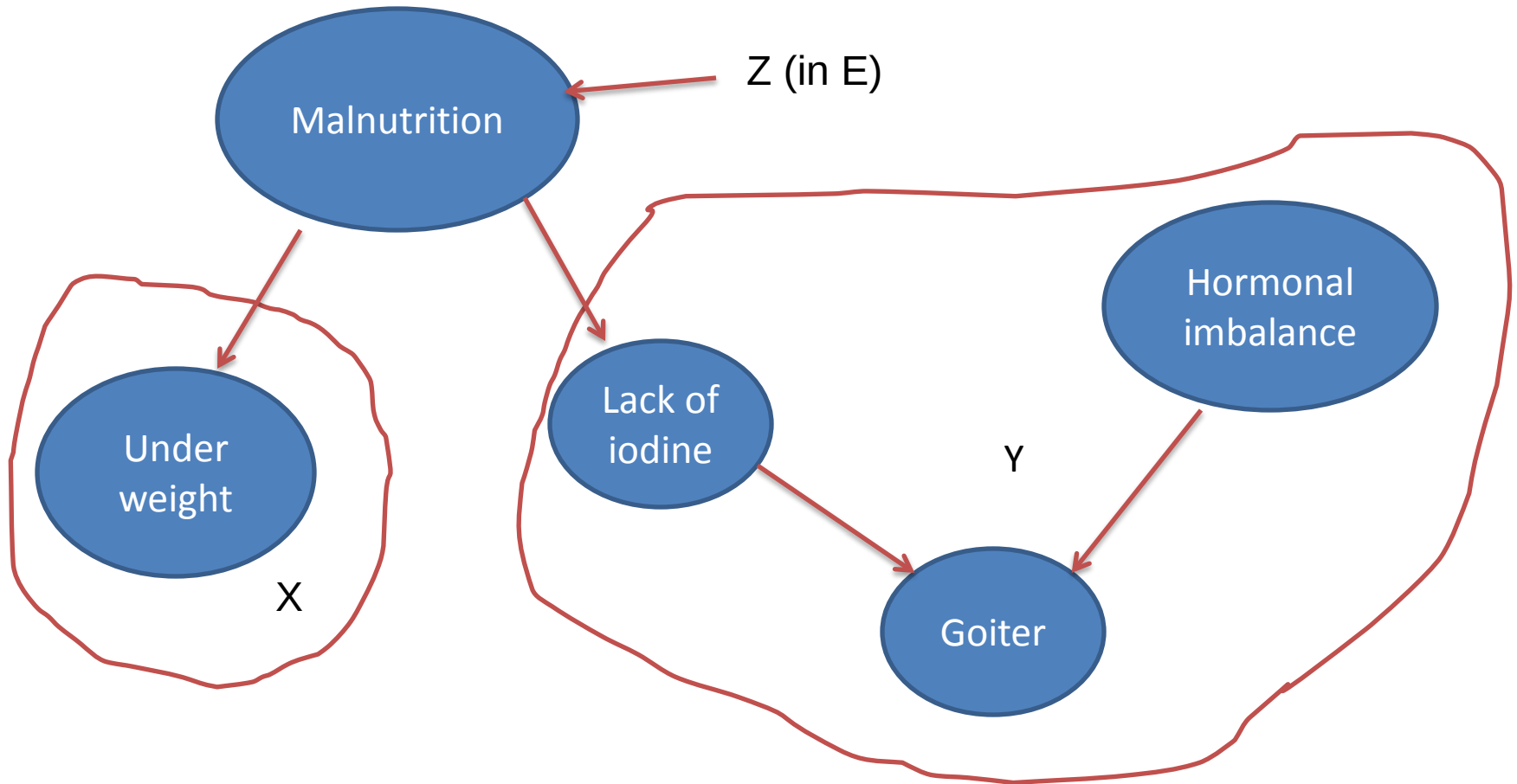
Conditional Independence Relation

- If every undirected path from the nodes in X and nodes in Y is d-separated by a given set of evidence nodes E , then X and Y are conditional independent given E
- A set of nodes E , d-seperates two set of nodes X and Y if every undirected path from a node in X to a node in Y is blocked given E

Example of d-separation



Example of d-separation



Conditional Independence relation

- A path is blocked given a set of nodes E if there exists a node Z in the path for which one of the following holds:
 - Z is in E and Z has one arrow on the path leading in and one leading out (Lack of Iodine)
 - Z is in E and Z has both path arrows leading out (Malnutrition)
 - Neither Z nor its descendants are in E and both path arrows lead into Z (Goiter)

Construction of Belief Networks

- Choose the set of variables X , that describe the ordering appropriately
- Choose an ordering for the variables (very important step)
- While there are variables
 - Pick a variable and a node for it
 - Set Parent (X) to some minimal set of existing nodes such that conditional independence is satisfied (lack of iodine)
 - Define probability table

Minimalist principle

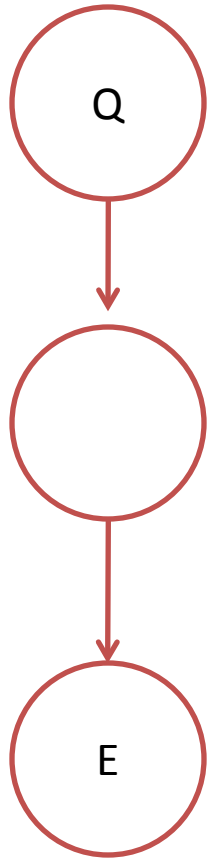
- Minimalist principle: a smaller explanation should be preferred rather than increasing the number of premises
 - In case of BN: the smaller the network, the better it is (both from explanatory and computational aspects)
 - Choose several ordering of the variables, construct the BN incrementally, keep the one with smallest number of edges

Inferencing in Bn

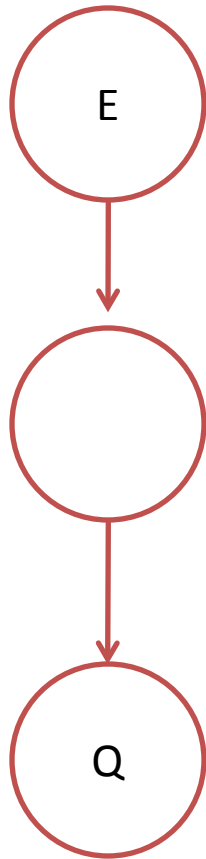
- Diagnostic inference (from effects to causes)
 - Given that Jhon calls, what is the probability of burglary? i.e. Find $P(B|J)$
- Causal inference (from causes to effects)
 - Given Burglary, what is the probability that
 - Jhon Calls, i.e $P(J|B)$
 - Mary calls, i.e $P(M|B)$

- Intercausal inference (between causes of a common event)
 - Given alarm, what is the probability of burglary?
i.e. $P(B|A)$
 - Now given Earthquake, what is the probability of burglary? i.e. $P(B|A \wedge E)$
- Mixed inferences (some causes and some effects known)
 - Given John calls and Earthquake, what is the probability of Alarm i.e. $P(A|J \wedge \neg E)$

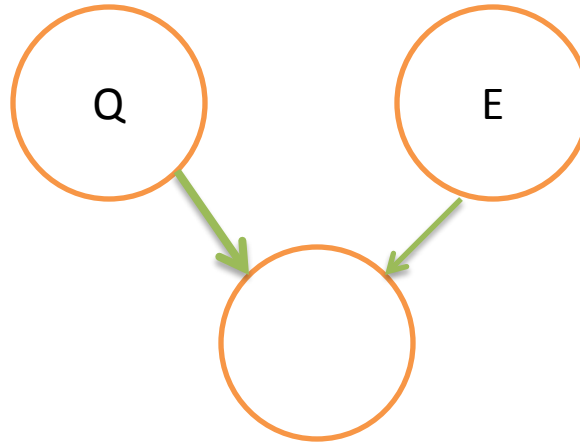
The four patterns



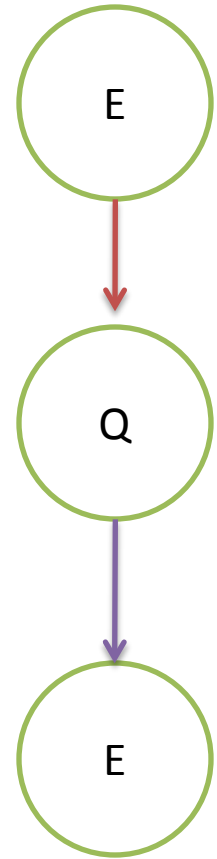
Diagnostic



Causal



Inter-causal

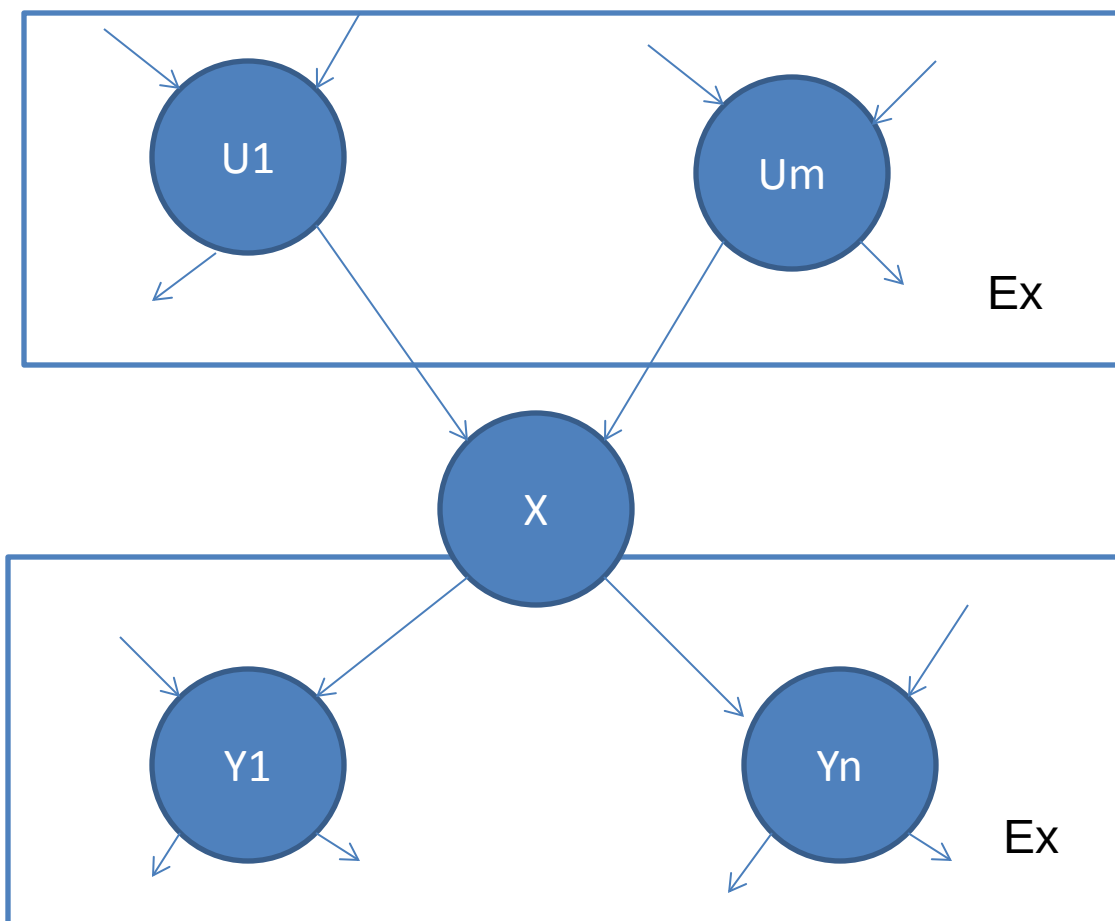


Mixed

- Calculate (Assignment)

Inferencing in Ploytree

- U: $U_1 \dots U_m$, parents of node X
- Y: $Y_1 \dots Y_n$, children of node X
- X: query variable
- E: evidence variables (whose truth values are known)
- Objective: compute $P(X|E)$



- Ex is the set of causal support for X
 - Variables above X connected through its parents which are known
- Ex is the set of evidential support for X
 - Variables below X connected through its children

- Computing $P(X|E)$
- Since X d-seperates E_x and E_x therefore, we can simplify the numerator as
- When $\quad$ is the constant representation denominator

- Both the term
- Can be computed recursively using the conditional independence relations
- If the parents are known, X is conditionally independent from all other nodes in the causal support set
- Similarly, given the children, X is independent from all other variables in the evidential support set

Joint Probability distribution

Ordering

- Example

Incremental Network Construction

- Choose the set of relevant variables X_i that describe the domain
- Choose an ordering for the variables (very important step)
- While there are variables left:
 - a) pick a variable X and add a node for it
 - b) set Parents (X) to some minimal set existing nodes such that the conditional independence property is satisfied
 - Define the conditional prob table for

Conditional independence relation

- D-separation

- Bayesian network corresponds to a factorization of the joint probability distribution
 - $P(A,B,C,D,E) = p(A)p(B) p(C|A,B)...$

- Factor graph
- Undirected graph
- Directed acyclic graph

- Belief networks represent conditional independence

Assignment

- **Application of a Bayesian Network in a GIS based Decision Making System**
- **Bayesian Networks for Reliability Analysis of Complex Systems**

Read the above papers and submit a report

Assignment

- Book
- Bayesian Networks and Influence Diagrams
- Read Chapter 1,
- Read Chapter 2

Chapter 2

- Causality
 - Serial connection
 - Diverging connection
 - Converging connection
 - Inter causal inferencing
 - The importance of correct modelling of causality
 - D-seperation criterion
 - Exercise

Chapter 3 (Probabilities)

- Probabilistic networks have a qualitative aspect and a corresponding quantitative aspect, where the qualitative aspect is given by a graphical structure in the form of an acyclic, directed graph (DAG) that represents the (conditional) dependence and independence properties of a joint probability distribution defined over a set of variables that are indexed by the vertices of the DAG.
- Axioms of probability
- Baye's rule

Chapter 4: Probabilistic Network

- Bayesian network (4.1)
- Influence diagram (4.2)
- Object-oriented probabilistic network (4.3)
 - Qualitative aspect is represented in terms of conditional dependence and independence
 - Quantitative aspect uses the probabilistic theories to assess the parameters that appears in the graphical model

Chapter 4: Reasoning under uncertainty

- A probabilistic interaction model between a set of random variables may be represented as a joint probability distribution.
 - Considering the case where random variables are discrete, it is obvious that the size of the joint probability distribution will grow exponentially with the number of variables as the joint distribution must contain one probability for each configuration of the random variables.
 - Therefore, we need a more compact representation for reasoning about the state of large and complex systems involving a large number of variables.
 - To facilitate an efficient representation of a large and complex domain with many random variables, the framework of Bayesian networks uses a graphical representation to encode dependence and independence relations among the random variables.

Chapter 4: Discrete Bayesian Networks

A (discrete) Bayesian network $N = (X, G, P)$ consists of

- a DAG $G = (V, E)$ with nodes $V = \{v_1, \dots, v_n\}$ and directed links E*
- a set of discrete random variables, X , represented by the nodes of G*
- a set of conditional probability distributions, P , containing one distribution,*

$P(X_v | X_{pa(v)}),$ for each random variable $X_v \in X$.

A Bayesian network encodes a joint probability distribution over a set of random variables, X , of a problem domain. The set of conditional probability distributions, P , specifies a multiplicative factorization of the joint probability distribution over X as represented by the chain rule of Bayesian networks):

Chapter 4

- Consult example 4.1 and 4.2
- Conditional linear Gaussian Bayesian Network
 - Consult example 4.3 and 4.4
- Differentiate among Bayesian Networks, Influence Diagram and Object-oriented probabilistic networks. Which one you consider best in taking decision under uncertainty?
- Solve the exercise

Chapter 5: Solving Probabilistic Networks

- In the case of probabilistic networks, we have a clear distinction between the knowledge base and the inference engine.
 - The knowledge base is the Bayesian network, whereas the inference engine consists of a set of generic methods to compute solutions to queries against the knowledge base.
 - Probabilistic inference in Bayesian networks consists of computing posterior beliefs in the light of evidence or KB.

Chapter 5: Inference approaches

- We consider variable elimination, query-based inference, arc reversal, and message passing in junction trees
- **Inference in Discrete Bayesian Networks**
 - One particular type of probabilistic inference task in Bayesian networks is the task of computing the posterior marginal of an unobserved variable Y given a (possibly empty) set of evidence ε , i.e., $P(Y \mid \varepsilon)$.

Chapter 6: Eliciting the model

- Depending on the nature of the problem, probabilistic networks might or might not be a natural choice of modeling methodology
- A probabilistic network consists of two components: structure and parameters (i.e., conditional probabilities)
 - The structure of a probabilistic network is often referred to as the qualitative part of the network, whereas the parameters are often referred to as its quantitative part

- Two key problems need to be addressed in the process of establishing the model structure: identification of the relevant variables and identification of the links between the variables.

Chapter 7: Modelling Techniques

Chapter 8: Data Driven Technique